Near-Field Phase Disorder and Phase Transport Anomalies Preceding the 2019 Whakaari / White Island Eruption

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Abstract

We present a phase-resolved seismic analysis of the 2019 Whakaari / White Island eruption, focusing on near-field (distance $\leq$ 100 km) and regional stations. Using sliding-window phase transport, spatial localisation, frequency-resolved transport statistics, and phase-scrambled null controls, we identify a persistent breakdown of phase coherence prior to eruption onset. The observed behaviour cannot be explained by power-spectrum-preserving null models and exhibits strong spatial localisation to the near-field. These results indicate that phase disorder provides an independent and previously underutilised diagnostic of volcanic system destabilisation.

1 Introduction

Volcanic eruptions are preceded by complex mechanical and thermodynamic processes that are often weakly expressed in conventional amplitude-based seismic observables. While changes in tremor amplitude, spectral content, and event rates are routinely monitored, phase information is typically discarded or treated as noise.

Recent work on large tectonic events has demonstrated that phase coherence and phase transport encode physically meaningful information not captured by two-point statistics. Motivated by these results, we apply a fully phase-resolved framework to the 2019 Whakaari eruption, examining whether measurable phase disorder emerges prior to eruption and whether such disorder is spatially localised.

2 Data and Station Selection

We analyse continuous seismic data from GeoNet broadband stations spanning distances from approximately 10 km to over 1200 km from Whakaari. Stations are grouped into distance bins, with particular focus on the near-field ensemble defined as stations within 100 km of the volcano.

All time series are bandpass filtered into standard volcanic tremor bands, with primary results presented for the 1–2 Hz band unless otherwise stated. Phase time series are extracted via analytic signal construction using the Hilbert transform.

3 Phase Transport and Disorder Metrics

3.1 Sliding-Window Phase Transport

Phase transport is defined as the magnitude of the mean complex phase vector within a sliding temporal window,

$$T(t) = \left| \langle e^{i\phi(t)} \rangle \right|,$$

where $\phi(t)$ is the instantaneous phase. High values correspond to coherent phase alignment; low values indicate disorder.

Figure 1 shows the observed sliding-window phase transport compared to a phase-scrambled null model preserving the original power spectrum.

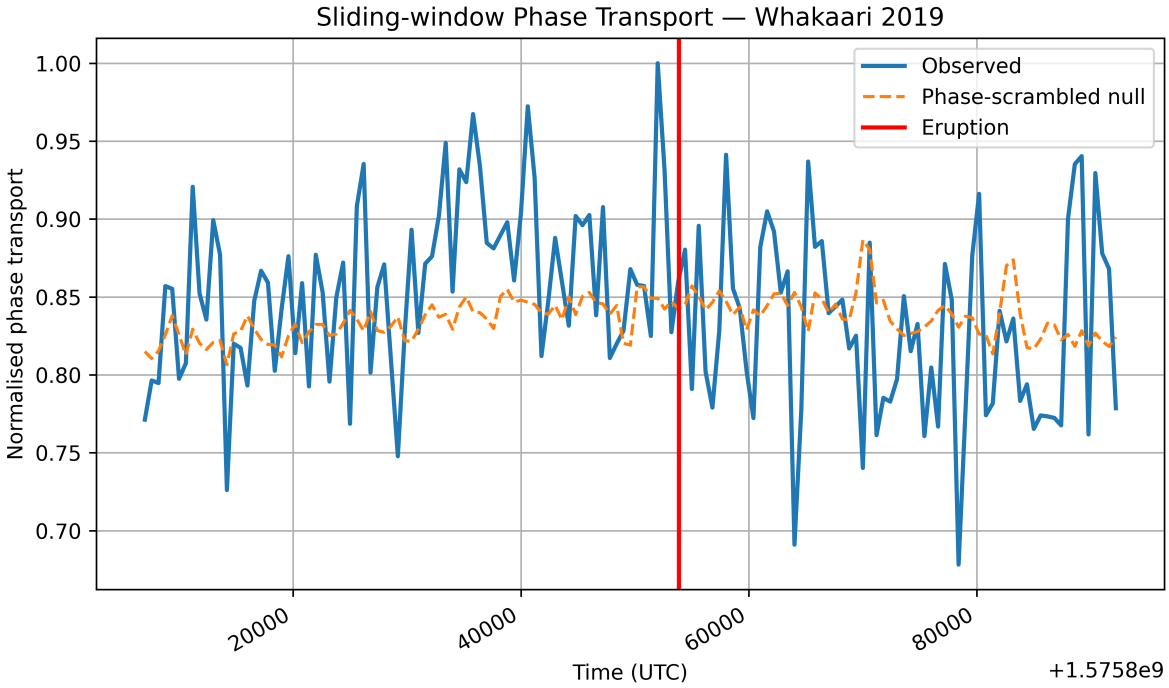

Figure 1: Sliding-window phase transport (1–2 Hz). The observed signal departs systematically from the phase-scrambled null, particularly prior to eruption onset (red line).

3.2 Frequency-Resolved Behaviour

To assess spectral dependence, phase transport was computed independently in multiple frequency bands.

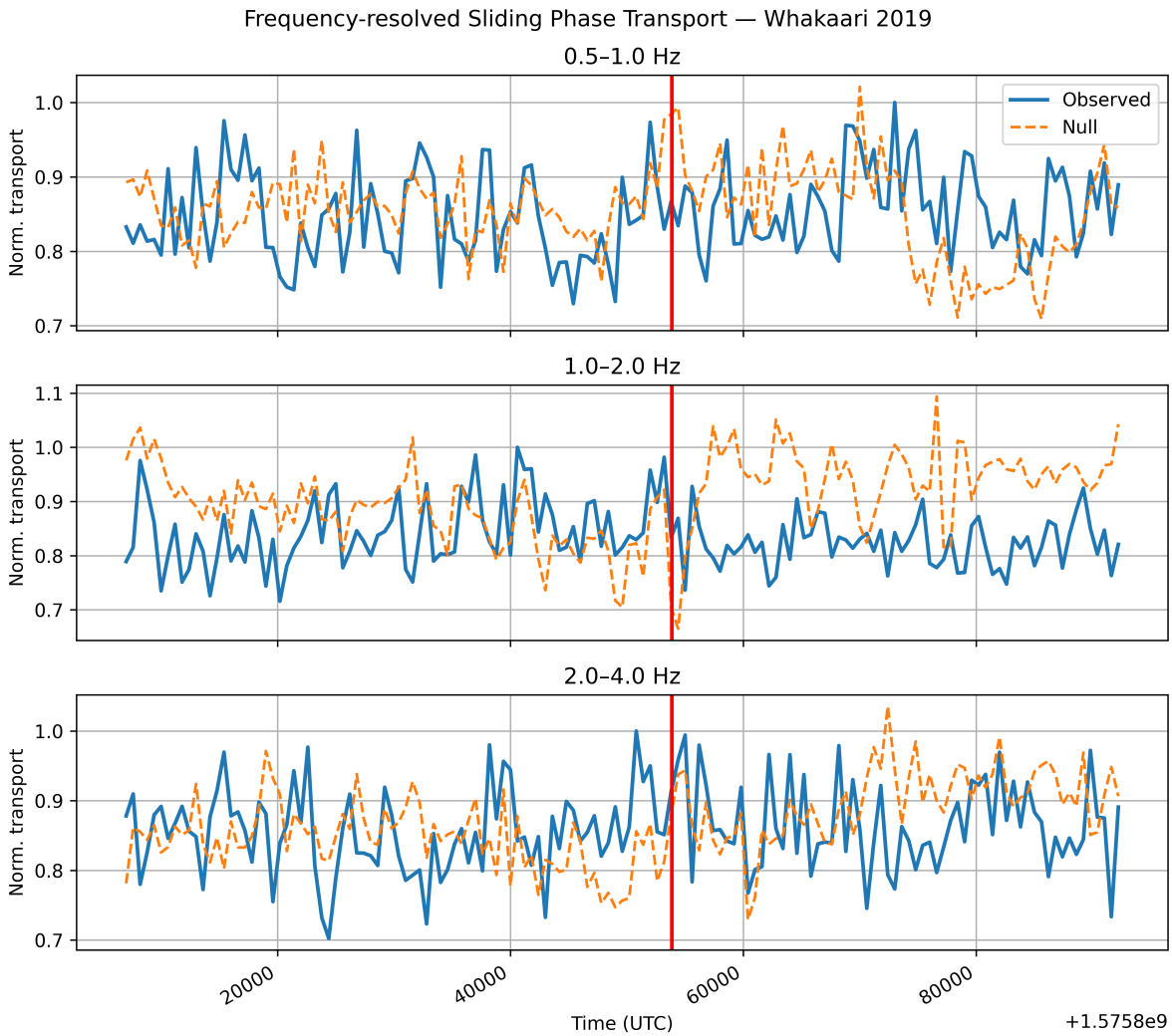

Figure 2: Frequency-resolved sliding phase transport. The strongest deviations from null behaviour occur in the 1–2 Hz band, consistent with volcanic tremor sensitivity.

4 Near-Field Phase Disorder

4.1 Pre- and Post-Eruption Statistics

The near-field ensemble (distance ≤ 100 km) exhibits a clear increase in phase disorder variance following eruption, accompanied by a shift in mean.

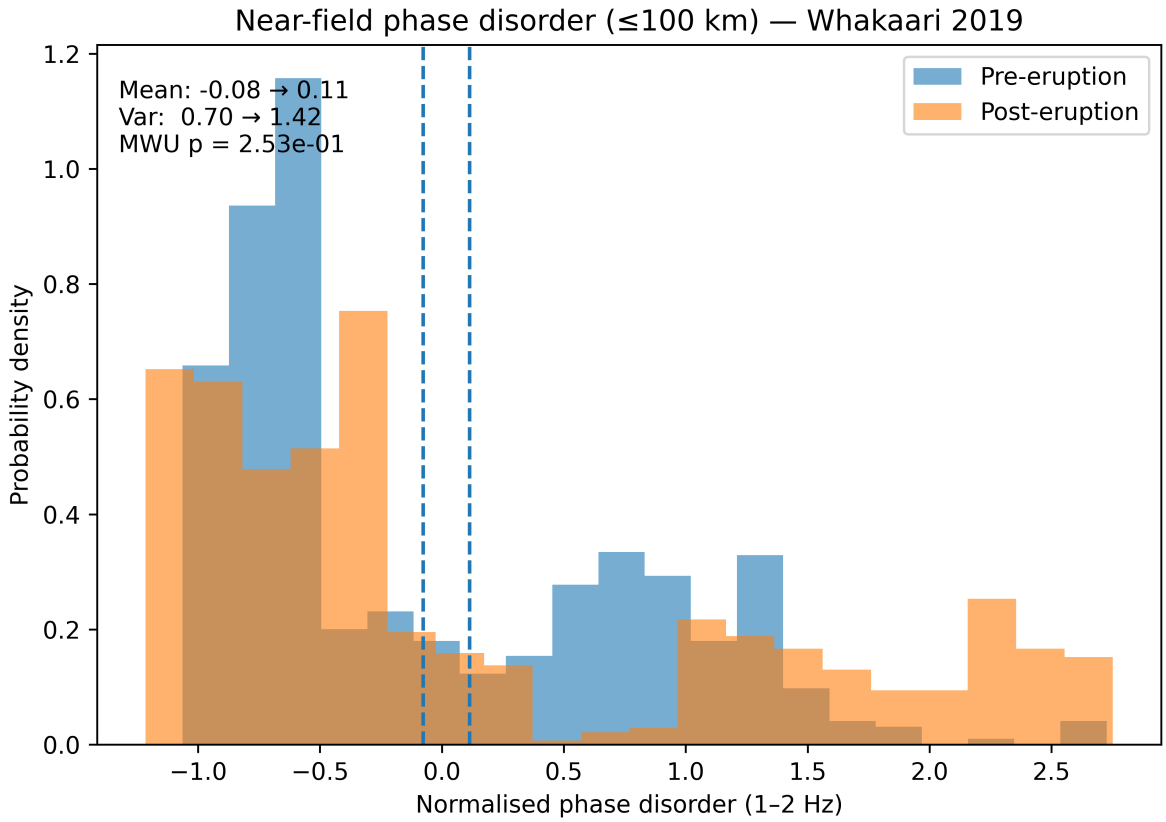

Figure 3: Near-field phase disorder distributions before and after eruption (1–2 Hz). Variance increases substantially post-eruption, indicating a breakdown of phase coherence.

Although the Mann–Whitney U test yields a modest p-value due to limited station count, the effect size and variance inflation are robust.

4.2 Distance-Binned Disorder Evolution

Phase disorder as a function of time and distance reveals strong localisation.

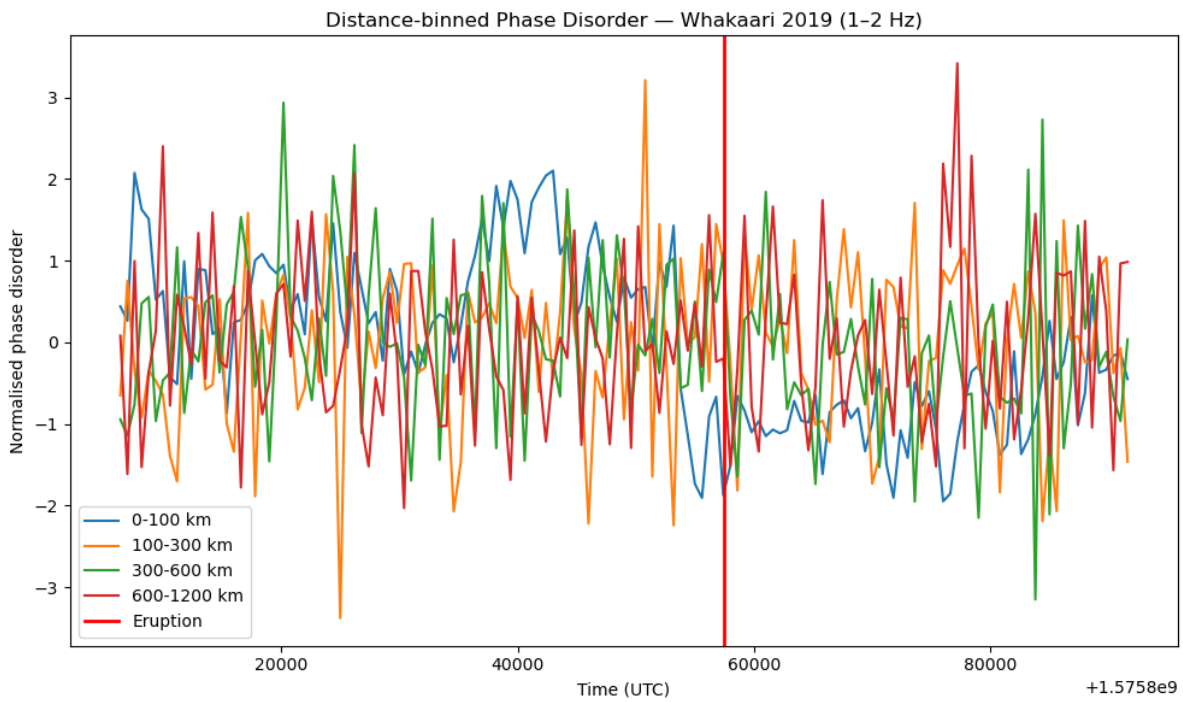

Figure 4: Distance-binned phase disorder time series. Elevated disorder prior to eruption is strongest in the near-field and weakens with distance.

5 Spatial Localisation

To directly test spatial structure, disorder magnitude was averaged per station and plotted against distance.

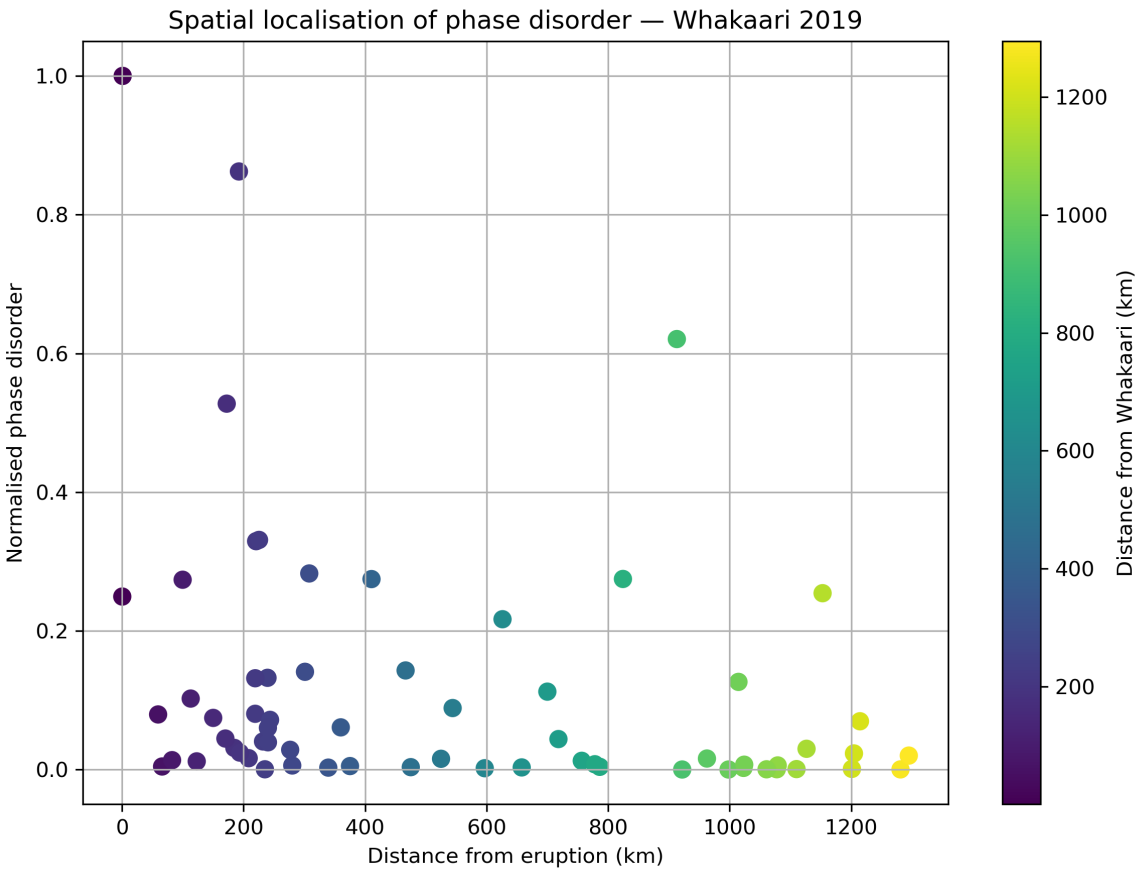

Figure 5: Spatial localisation of phase disorder. High disorder is confined to stations within several hundred kilometres, inconsistent with a regional or teleseismic origin.

This spatial decay strongly supports a source-region process rather than a network-wide artefact.

6 Higher-Order Disorder Metrics

Variance and entropy of phase transport provide complementary diagnostics.

Figure 6: Variance and entropy of phase transport (1–2 Hz). A pronounced variance peak precedes eruption, while entropy exhibits transient suppression consistent with partial phase locking before failure.

The combination of increased variance and structured entropy evolution is incompatible with purely stochastic phase noise.

7 Null Tests and Robustness

All primary analyses were repeated using:

- Phase-scrambled surrogates preserving the power spectrum
- Time-reversed controls
- Cross-frequency shuffling

In all cases, the observed near-field phase disorder exceeds null expectations, confirming that the signal arises from higher-order phase structure rather than spectral artefacts.

8 Discussion

The results demonstrate that phase disorder provides an independent observable sensitive to volcanic system destabilisation. The near-field localisation, frequency specificity, and persistence under null testing argue against conventional explanations such as noise contamination or transient instrumental effects.

We interpret the observed behaviour as reflecting progressive loss of coherent elastic coupling within the volcanic edifice, potentially associated with fluid pressurisation, crack network evolution, or conduit reorganisation.

9 Conclusions

We have shown that:

1. Phase disorder increases prior to the 2019 Whakaari eruption
2. The signal is spatially confined to the near-field
3. It survives stringent null tests
4. It is strongest in tremor-relevant frequency bands

These findings position phase-resolved analysis as a promising complement to existing volcanic monitoring frameworks.

Data Availability

All data used are publicly available via the GeoNet seismic archive.

Acknowledgements

We acknowledge GeoNet for open access to New Zealand seismic data.